Weekly Progress Report

Refactoring Detection Project - Week 8 progress update for the Monday research supervisor meeting

1. Overview

During Week 8, the team moved closer to the final delivery stage of the CLONE interface work. The main focus was finishing the last UI bugs, expanding the Overleaf research paper, preparing clearer documentation for the plugin workflow, and organizing materials for user testing. The team showed the UI fixes completed so far and confirmed that the project website and shared files remained up to date.

A major part of the week was dedicated to the paper and final documentation. The team added two important paper sections, Workflow and How to Use, added more detail to the existing sections, and prepared light-mode images of the CLONE UI features, including individual feature screenshots and one full UI panel image. Dr. AlOmar also shared a reference paper that can help guide the final writing, and the team discussed collecting structured user feedback through a testing form.

2. Weekly Progress Summary

Area	Progress Made
UI fixes	Finished the remaining bug-fix work that was blocking the final panel review, with the goal of verifying the final three UI issues in the full IntelliJ environment.
Overleaf	Continued writing and expanding the final research paper, including more detail in the abstract, introduction, problem statement, new features, workflow, and usage sections.
New sections	Added two major sections to the paper: a Workflow section explaining how CLONE works and a How to Use section explaining installation, configuration, and plugin usage.
Images and figures	Added feature images in IntelliJ light mode, including screenshots of individual CLONE UI sections and one image showing the full UI panel.
Website and files	Kept the team website current and uploaded the weekly files to OneDrive so the project documentation remains organized in one place.

3. Meeting Discussion

The Week 8 meeting focused on finalizing the work needed for the paper, demo, and user testing. The team showed the UI fixes and reviewed the progress already completed in Overleaf. Dr. AlOmar shared a paper that can be used as a reference while improving the final report. The team also discussed how to find users to test the CLONE plugin with the updated panel, and Dr. AlOmar provided a form structure that testers can use to submit feedback.

Topic	Discussion / Decision
UI fixes	The team showed the latest UI fixes and confirmed that the last bugs were being completed and prepared for verification.
Overleaf paper	The paper continued to grow with added content, new sections, and screenshots documenting the CLONE workflow and user-facing features.
Reference paper	Dr. AlOmar shared a paper that may help the team improve the structure, framing, and evidence used in the final research paper.
User testing	The team discussed finding users to test the plugin and collecting feedback through a structured form that can later support graphs and experimental results.

4. Overleaf and Final Paper Progress

The Overleaf work became more complete during Week 8. The team added a Workflow section to explain the end-to-end behavior of CLONE, including how duplicate Java code is detected, how the proposed Extract Method refactoring is generated, how the panel presents validation evidence, and how the developer decides whether to apply the change.

The team also added a How to Use section. This section explains that CLONE is distributed as an IntelliJ plugin, how the plugin is installed from a ZIP file, how it is configured in IntelliJ settings, how model/provider information is entered, and how copy-and-paste actions trigger the CLONE workflow. This section is important because it gives readers and testers a practical path for running the tool.

The New Features content was expanded because some subsections were still too brief. Each feature section now needs enough explanation to connect the interface element to the developer decision it supports. The planned screenshots help show the full UI panel, the explanation area, clone occurrences, diff preview, AI Trust/validation, action buttons, and related details.

5. UI Documentation and Screenshots

Documentation Item	Purpose
Full UI panel image	Shows how the main CLONE interface brings explanation, diff preview, AI Trust, details/provenance, and action buttons together in one panel.
Individual feature screenshots	Supports each New Features subsection by showing the exact part of the panel being described.
Light-mode images	Keeps the final paper visuals readable and consistent with IntelliJ documentation style.
Workflow description	Explains how the plugin moves from copy-paste detection to proposal review, validation evidence, and developer action.
How-to-use instructions	Explains installation, model/provider setup, token requirements, and the recommended small duplicated code example for testing.

6. User Testing Plan

The team began organizing user testing for the final CLONE panel. Each team member is expected to find two to three users who can test the plugin and fill out the feedback form. The form should collect information that can be converted into graphs for the final paper, such as overall experience, most useful feature, clarity of the diff preview, trust in validation evidence, and ease of using action controls.

The testing instructions should ask users to use a small and simple duplicated Java code example because the plugin uses API tokens during detection, refactoring, validation, and testing. If the selected model provider does not have available tokens or quota, the workflow may not run completely, and users should report that behavior in the form.

7. Week 8 Tasks

Task	Expected Outcome
Fix final 3 UI issues	Complete the final panel fixes and verify them by running the updated plugin in IntelliJ.
Expand Overleaf subsections	Add more detail to each New Features subsection so the paper explains what each feature does and why it matters.
Add Workflow section	Explain the full CLONE process from copy/paste detection to review, validation, and Apply/Cancel decisions.
Add How to Use section	Document plugin installation, configuration, model/provider setup, API key/token needs, and a simple test workflow.
Add feature images	Place images of each new feature in the correct subsection, using IntelliJ light mode for readability.
Recruit testers	Each team member should find two to three users to test CLONE and complete the feedback form.
Edit feedback form	Improve the form questions so the answers can be summarized into graphs for the Experimental Results section.

8. Challenges and Blockers

The main challenge for Week 8 was moving from implementation work into final verification and evidence collection. The UI fixes need to be fully checked in IntelliJ, and the Overleaf paper needs enough detail to explain the purpose of each interface feature clearly. The team also needs to collect user feedback quickly enough to analyze responses and add graphs to the final report.

Another constraint is the token/model setup needed to run the plugin completely. Since CLONE uses AI model calls as part of its workflow, a user without available tokens or a valid API setup may not see the full detection, refactoring, validation, and testing sequence. This needs to be documented in the user-testing instructions and considered when interpreting test feedback.

9. Immediate Next Steps

- Finish and verify the final three UI issues in the full CLONE panel.
- Complete the Overleaf paper, especially the Workflow, How to Use, and New Features sections.
- Add all individual feature screenshots and the full UI-panel image to the paper.
- Finalize the Google Form so it captures useful feedback for graphs and experimental results.
- Have each team member recruit two to three testers and collect their responses.
- Summarize user feedback with charts, including overall experience and most useful feature.
- Keep the project website updated with the final report, videos, UI screenshots, and weekly documents.

10. Conclusion

Week 8 focused on transitioning the CLONE work into final documentation, testing, and reporting. The team finished the latest UI fixes, expanded the Overleaf paper, added the Workflow and How to Use sections, prepared feature images, and planned user testing with a feedback form. The next stage is to complete the final UI verification, gather tester feedback, convert the responses into report graphs, and finish the final paper and website deliverables.